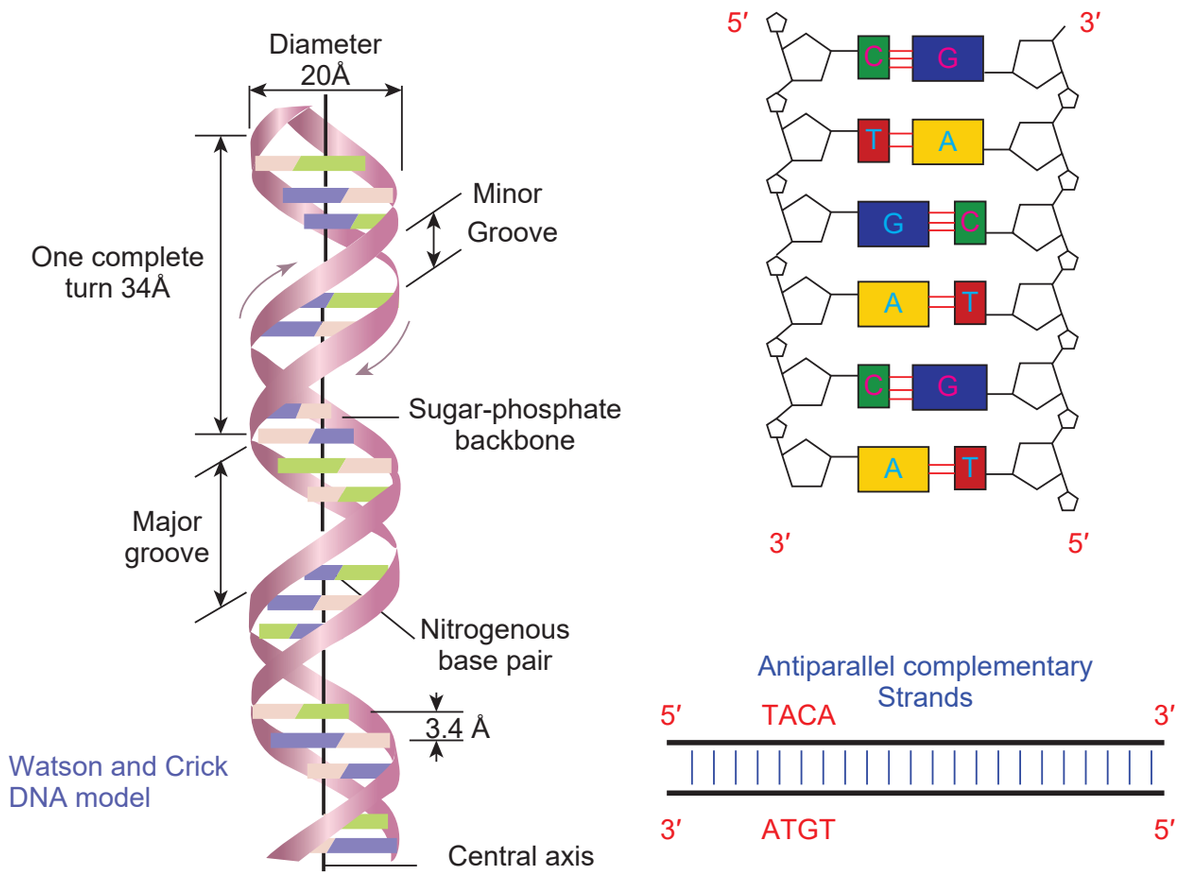


Figure 14.9: Watson and Crick DNA model, Antiparallel dsDNA

important in the process of DNA replication and gene expression.

- The pairing of two strands creates a major groove and minor groove on the surface of the duplex.
- The two strands are antiparallel, that is their 5′, 3′ phosphodiester bonds run in opposite directions.
- The vertically stacked bases are 3.4 A° apart.
- Each complete turn of double helix contain base pairs, which is 34 A° units long.

14.2.2 Erwin Chargaff's Rule

In the late 1940s Erwin Chargaff and his colleagues found that the four nucleotide bases of DNA occur in different ratios in the DNA of different organisms. Erwin Chargaff measured the quantity of the bases in DNA and noticed that the number of Adenine is equal to the number of Thymine and the number of Guanine is equal to the number of Cytosine residues. Hence the sum of Purine residues equal to the sum of the Pyrimidine residues.

Quantitatively A=T or A/T = 1
C=G or C/G = 1
A+G = T+C

HOTS

If percentage of adenine in one of the DNA strand is 20. Can you determine the percentage of other bases. If yes how?

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